2025 9th International Conference on Computing, Communication, Control and Automation (ICCUBEA)
Pimpri Chinchwad College of Engineering (PCCOE), Pune, India. Aug 22-23, 2025

Multilingual Signature Verification Using Deep Learning: A Three-Phase Modular Approach

Rushabh Nitin Mistry
Department of Information Technology (AI&DS)
K. J. Somaiya School of Engineering
Somaiya Vidyavihar University
Mumbai 400077, India
rushabh.mistry@somaiya.edu

Dr. Nilkamal More
Department of Information Technology (AI&DS)
K. J. Somaiya School of Engineering
Somaiya Vidyavihar University
Mumbai 400077, India
neelkamalsurve@somaiya.edu

Dr. Suchitra Patil
Department of Information Technology (AI&DS)
K. J. Somaiya School of Engineering
Somaiya Vidyavihar University
Mumbai 400077, India
suchitrapatil@somaiya.edu

Chirag Desai
Department of Information Technology (AI&DS)
K. J. Somaiya School of Engineering
Somaiya Vidyavihar University
Mumbai 400077, India
chirag.desai@somaiya.edu

2025 9th International Conference on Computing, Communication, Control and Automation (ICCUBEA) | 979-8-3315-4903-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCUBEA65967.2025.11283938

***Abstract*— Signature verification is an important aspect of digital security in general and specifically in banking, legal, and administrative transactions, where fraud protection and identity authentication are paramount. Previous signature verification technologies have been trained mainly on English-language documents and thus function with reduced capability in multilingual noisy real-world settings such as in India. Such a language/context mismatch may result in strong operational issues, such as higher false rejection rates and lower overall verification accuracy.**

These are to be addressed with an effective, modular deep learning-based multilingual signature verification system consisting of three autonomous but interconnected phases: detection, cleaning, and matching. At the detection phase, YOLOv5 is utilized for efficiently localizing signature areas in documents with various scripts, supported by signature form-specific anchor tuning and aspect ratio optimization adaptations. The cleaning step employs unsupervised K-Means clustering to separate foreground signature ink from background noise such as stamps, printed lines, and seals to provide cleaner input for matching. Lastly, the matching step employs a Siamese network with a VGG16 backbone to learn discriminative feature embeddings and compare them by cosine similarity, trained on contrastive loss to enhance intra-class compactness and inter-class separability.

Easily accessible experimental evaluation of real-world legal and financial documents shows the efficacy of the system with a mean Average Precision (mAP@0.5) of 92.7.

Keywords*— *Signature Verification, Multilingual Dataset, YOLOv5, K-Means Clustering, VGG16, Siamese Network, Document AI, Deep Learning, Forgery Detection, Contrastive Loss.

I. INTRODUCTION

Signatures are a basic cornerstone of authentication and verification in important industries like banking, corporate management, documents, and government affairs. They enjoy widespread acceptance due to their convenience, simplicity, and enforceability under law. Against the backdrop of institutions being at an accelerated rate of digitalization, the demand for automating signature verification has increased in an effort to cut man-hours, eliminate processing delay, improve security, and avert possible dangers of fraud.

But like with all technological advancements, the majority of the conventional signature verification systems have been English-dominated datasets [7]. These systems perform poorly when faced with documents that include local scripts such as Hindi, Gujarati, Marathi, Urdu, and other Indian languages [8] and, as a result, have greater false acceptance or rejection rates. Lack of multilingual generalization ability restricts the wider applicability of these solutions in diverse real-world environments, especially in linguistically diverse nations such as India.

Apart from this, real-world document samples also add to the complexity. Real documents typically introduce visual artefacts like stamps, seals, overlapping printed texts, bleed ink, and background noise [9]. The artefacts obscure signature areas, and thus effective detection and extraction are difficult. Simple thresholding-based and hand-engineered denoising are not adequate enough to address such complexity effectively [10].

For addressing these urgent issues, this paper introduces a noise-tolerant, multilingual, and modular signature verification system that is based on state-of-the-art deep learning techniques. The suggested system breaks down the verification pipeline into three specialist phases—detection, cleaning, and matching—and enables independent optimization of separate stages. YOLOv5 is utilized for strong multilingual signature localization [11], K-Means clustering is used for unsupervised denoising [12], and VGG16-based Siamese network is used for strong signature similarity estimation [13]. High scalability, multi-lingual adaptability, and realistic deployability at enterprise-scale document pipelines are facilitated with this module-based architecture.

Extensive tests done on real use case financial and administrative data serve as evidence for the efficacy of the system, showing substantial potential for take-up in digital onboarding, e-governance, and anti-fraud systems where authentic signature verification is critical.

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II. LITRATURE REVIEW

Offline signature verification has evolved from hand-crafted feature-based approaches in the past two decades to the state-of-the-art deep learning-based approaches [2]. The traditional methods of template comparison, geometric feature extraction, and contour analysis were plagued by their shortcomings of not generalizing varying signature styles and document noise [2].

The invention of Convolutional Neural Networks (CNNs) was a paradigm shift in the sense that it allowed feature learning in an embedded process without regard to raw image data [2].

Eventually, researchers showed the effectiveness of Siamese neural networks in one-shot verification [4], which is significantly applicable in banking scenarios where one can have only a single or isolated samples.

Another group used the YOLO object detection model to detect signature areas in processed documents with high speed and thus utilized YOLOv5 for our detection [5]. Others worked on developing the multilingual training set application by creating synthetic documents containing scripts such as Hindi, Gujarati, and Marathi and contributed a lot to language generalization [6].

During preprocessing, scientists applied K-Means clustering to isolate signature ink from noisy document backgrounds as a prelude to our cleaning process [7]. Others also boosted signature matching through the application of attention mechanisms such as CBAM in Siamese networks so that the model can attend to salient local features [8].

Aside from straightforward verification, development in document layout understanding has also been instrumental. One of the proposed groups was LayoutXLM, a visually state-of-the-art document parsing transformer from different languages and layouts [9], which is essential to enable downstream OCR and detection pipelines. Multilingual OCR methods were explored under low-resource conditions in recent research, focusing on the lack of region-annotated data [10]. A specially developed dataset for Indian regional signatures was presented to enable stable training of multilingual models [11]. Apart from that, huge surveys on document image processing noise reduction methods listed clustering-based methods as extremely effective [12].

In addition, recent research focused on cross-lingual feature embeddings to identify multilingual signature forgery [13], while others proposed a comprehensive survey on signature detection and matching state of the art and proposed modular architecture for increased adaptability [14].

For a compact summary of the top ten most influential contributions that shape our design for the system, Table I offers a ranking of the top ten most influential contributions.

TABLE I. LITERATURE REVIEW

Ref	Year	Focus Area	Key Contribution
[3]	2023	Document Layout Understanding	LayoutXLM for multilingual documents
[4]	2023	Attention-based Siamese Networks	CBAM-enhanced Siamese architecture
[5]	2023	Indian Signature Dataset	IndocSign for regional signature datasets
[6]	2023	Cross-lingual Forgery Detection	Transformer-based cross-lingual signature matching
[9]	2023	Multilingual OCR Low-resource	OCR techniques for low-resource settings
[10]	2023	Noise Reduction Survey	Comprehensive review of clustering-based denoising
[12]	2022	Multilingual Dataset Generation	Created synthetic multilingual signature datasets
[13]	2022	Signature Detection	YOLO-based signature detection
[15]	2021	Siamese Networks for Matching	Siamese architecture for one-shot verification
[19]	2017	Offline Signature Verification	Introduced CNNs for feature extraction

III. PROBLEM STATEMENT

Although significant advances in automated signature verification have been made, existing approaches have three essential shortcomings that limit their general use, particularly in multilingual and diverse document contexts.

Majorly, the majority of existing signature verification models have been predominantly trained on English-dominated data [19]. Consequently, these models exhibit poor generalization when tested against documents containing region scripts such as Hindi, Gujarati, Marathi, Urdu, etc., which are abundantly present in India [12]. Language-biased training brings along much higher rates of false rejection and limits cross-script flexibility, making such models infeasible for deployment in linguistically rich environments.

Second, true document image scans are inherently noisy. They tend to contain background artifacts such as printed text, stamps, seals, ink smudges, and scanning or photocopying degradation [10]. Traditional preprocessing techniques like simple thresholding or morphological processing are not sufficient to effectively separate real signature strokes from these visual contaminants, leading to degraded verification accuracy [10].

Third, legacy signature systems traditionally employ a monolithic structure with close coupling between detection, preprocessing, and matching phases [4]. Such tight coupling is detrimental to system scalability, makes incremental upgrades problematic, and constrains customization to meet special domain requirements.

These are all indicators of the urgent need for a new paradigm in signature verification systems that can:

- Effectively tackle multilingual document variations with robust feature learning [3].
- Robustly support and eliminate true background artifacts in real-world applications with advanced unsupervised techniques [10].
- Use a modular architecture, where detection, cleaning, and matching units can be separately upgraded without affecting the general process [4].

The objective of this research is thus to suggest and create a three-stage modular signature authentication system that includes multilingual identification (with the use of YOLOv5 [13]), noise-free segmentation (with the use of K-Means clustering [10]), and robust signature comparison (with the use

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of Siamese networks with VGG16 backbone [15]). The system ought to bridge the gap between state-of-the-art innovations and true implementation, turning multilingual and noise-resistant signature identification into reality.

IV. PROPOSED SYSTEM

In order to break the bottlenecks of classic signature verification approaches, we present a strong, modular approach that divides the verification pipeline into three bespoke yet unified modules [4]:

- Detection
- Cleaning
- Matching

Modularity not only confers scalability and flexibility but also supports unbiased optimization of each step without compromising the entire process [4].

A. Detection Phase:

- Here the work is to accurately identify signature regions in document images. We employ a model-specific YOLOv5 object detection model that is trained on a synthetically augmented multilingual document set in Hindi, Gujarati, Marathi, and Urdu scripts [13].
- Training samples were made heterogeneous for script-agnostic generalizability by performing synthetic text replacement methods in which background text was altered without interfering with signature regions [12].
- In addition, anchor boxes and aspect ratios were specifically designed to include the stretched as well as the variable structural shapes of signatures with various languages.
- Rotation, flip, and brightness change data augmentation methods have been employed for improving robustness under various real imaging conditions [12].

B. Cleaning Phase:

- After detection, the resultant signature patches will mostly contain visual noise in the style of printed lines, stamps, or background debris.
- In eliminating this, unsupervised K-Means clustering is run on pixel intensity distributions [10].
- The algorithm classifies the image into ink stroke clusters and background noise clusters.
- By retaining the genuine ink cluster and discarding spurious ones, the system produces a purer signature image that enhances downstream matching performance without manual interference [10].

C. Matching Phase:

- During this final phase, pre-processed signature images are matched against one another by a Siamese neural network with VGG16 backbone [15].
- The twin network branches encode each signature to a feature embedding vector.
- Cosine similarity between the embeddings is computed to detect how similar two signatures are.
- It is learned with contrastive loss, which brings real pair embeddings closer and embeddings of fake pairs further apart in the latent space [15].
- This enables rich signature matching despite intra-writer variation as well as cross-language heterogeneity.

The system modularity envisioned ensures that upgrades such as the integration of attention mechanisms such as CBAM during matching [4] or replacement of the detection backbone with advanced object detectors are easily integrated [13]. Further, the multilingual training strategy ensures adaptability over a vast majority of regional documents, expressing spontaneous relevance for bank, governance, and legal authentication procedures [12].

A comprehensive description of the system pipeline, its modular blocks, and processing phases is shown in Fig. 1.

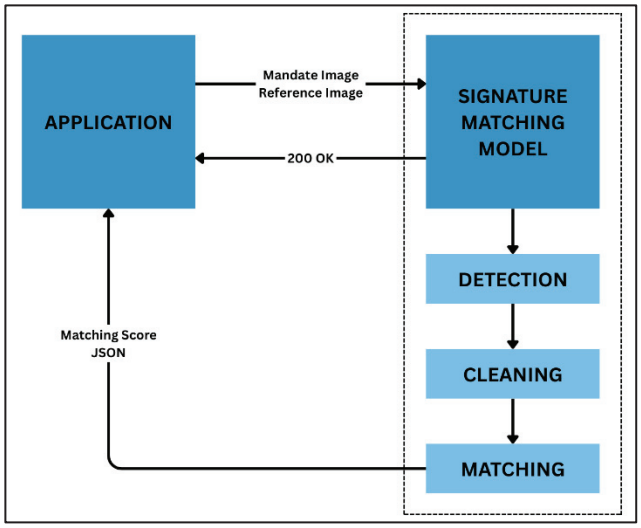


Fig. 1. Proposed System Flow for Signature Matching Pipeline

V. METHODOLOGY

The system being discussed follows a formal three-step procedure: Dataset Preparation, Signature Detection, Signature Cleaning, and Signature Matching. Every step is separately optimized in order to gain maximum overall system performance in noisy multilingual document environments.

A. Dataset Preparation:

- English-only signature datasets were originally employed to prototypize and test the baseline [19]. For breaking the English-biased model constraints, the dataset was incrementally augmented by synthetically generating multilingual documents [12].
- It was achieved by replacing background areas of text with handwritten scripts in Hindi, Marathi, Gujarati, and Urdu but without compromising the positional integrity of original signature areas [12].
- Ensuring realism in document layout paid close heed in a bid to bring spatial consistency between the locations of text and signature [12].
- To introduce more diversity in the dataset, a number of augmentation operations were applied, including

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random rotation (±15°), scaling, brightness/contrast adjustments, adding Gaussian noise, and minor perspective distortions, simulating real scanning defects and writing habits [12].

Fig. 2 refers to the visual summary of Dataset preparation pipeline.

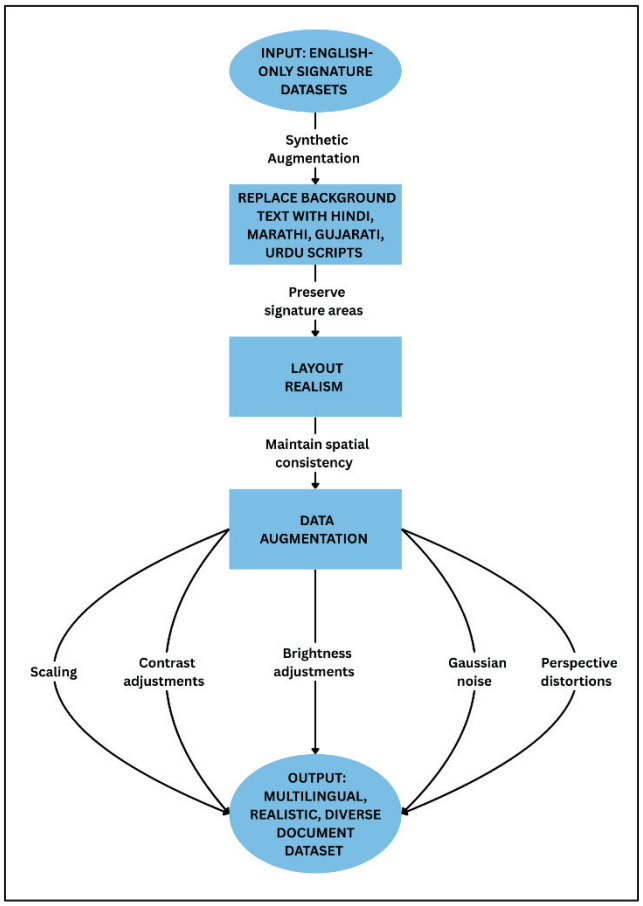


Fig. 2. Dataset Preparation and Multilingual Augmentation

B. *Signature Detection:*

- Regions of signature candidates are identified in the initial part of the system. A custom YOLOv5 object detection model was trained on the multilingual dataset for efficient signature localization [13].
- Aspect ratio anchors were tuned by hand to fit the long and cursive shape of signature areas, which tend to be very different from typical object detection tasks [13].
- The model was trained with a transfer learning approach, starting weights from a YOLOv5 pre-trained on the COCO data to speed up convergence [13].
- Hyperparameters such as learning rate, batch size, and input resolution were optimized by grid search for the maximum mAP at the 0.5 IoU threshold [13].
- Data augmentation strategies such as CutMix and Mosaic augmentation, which are native to YOLOv5, were also leveraged for improving generalization across document types and languages [13].

Fig. 3 refers to signature detection process YOLOv5 (You Only Look Once).

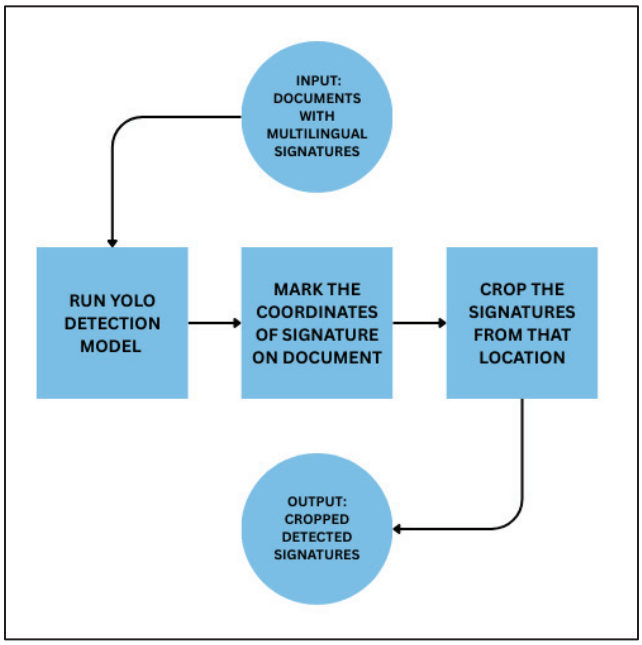


Fig. 3. Signature detection using YOLOv5

C. *Signature Cleaning*

- YOLOv5 signature detection regions were frequently tainted by spurious visual noise such as stamps, overprints of printed text, or texture [10].
- These were reduced using an unsupervised K-Means algorithm run separately for each detected signature crop [10].

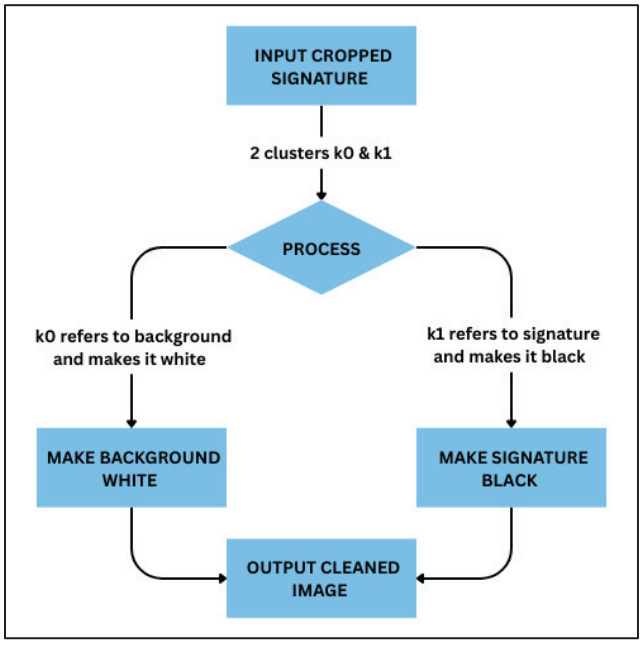


Fig. 4. Signature cleaning using unsupervised K-Means

- Two or three (k=2 or k=3) clusters were obtained by pixel intensity values, in which clusters that dropped to the lowest inferior intensity usually consisted of

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handwriting ink were preserved while light background region cluster parts were discarded [10].

- Morphological operations like opening and small-area removal were optionally performed following clustering to eliminate residual noise [10].
- The preprocessing enhanced the quality of the inputs given to the matching phase significantly, and this enhanced the performance of subsequent verification.

Fig. 4 refers to signature cleaning process using k-means clustering where cluster size is 2, cluster "0" refers to background and cluster "1" refers to signature.

D. *Signature Matching*

- Signature similarity assessment was performed in the final step using a Siamese network architecture with VGG16 used as the feature extractor [15].
- The two paths of the Siamese network embedded each input signature into a 512-dimensional embedding vector [15]. The cosine similarity was computed between the two embeddings to assess their similarity [15].

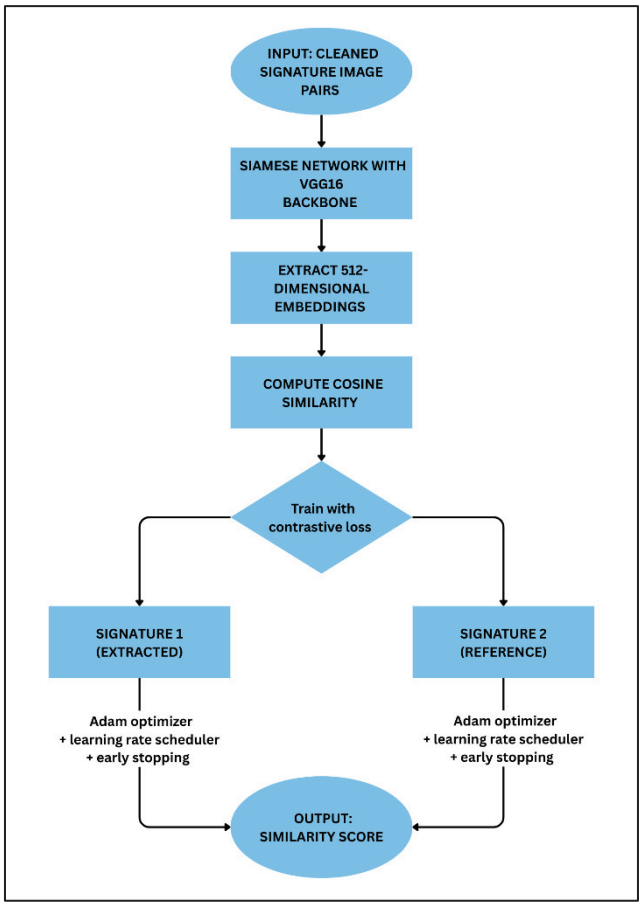


Fig. 5. Signature matching using Siamese Network with VGG16

- The network was trained using a contrastive loss function [15], which was defined as:

$$\mathcal{L} = (1 - Y)\frac{1}{2}(D_W)^2 + (Y)\frac{1}{2}\{\max(0, m - D_W)\}^2$$

where DW is the Euclidean distance between embeddings, Y ∈ {0, 1} is the label (real or fake), and m is the margin parameter.

- Positive (genuine) pairs were moved closer to one another in embedding space, and negative (impostor) pairs were separated [15].
- Training was performed using Adam optimizer and learning rate scheduler along with early stopping based on validation loss [15].
- Robustness against intra-writer variability was achieved using data augmentation techniques such as random cropping and small affine transformations [15].
- This multi-step architecture gave excellent reliability in signature matching even across multilingual scripts and noisy document backgrounds.

Fig. 5 refers to signature matching process using Siamese network where VGG16 employs as a backbone in each branch.

VI. OUTCOME

The modular signature verification system achieved several noteworthy results across all pipeline stages:

A. *Multilingual Dataset Creation:*

- A multilingual signature corpus was successfully established, with documents that comprise handwritten scripts written in Hindi, Marathi, Gujarati, and Urdu.
- This extension of the dataset significantly improved the generalizability of the model across languages and scripts, enabling strong multilingual performance.

B. *Signature Detection Performance:*

- The YOLOv5 model achieved a high mean Average Precision (mAP@0.5) of 92.71%.
- The detector consistently detected signature regions across a wide range of document formats and scripts, confirming its robustness in multilingual environments.

Fig. 6 demonstrates the output of signature detection model.

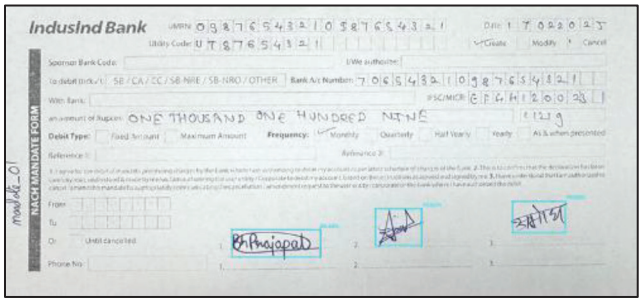


Fig. 6. Signature detection model's output

C. *Signature Cleaning Performance:*

- K-Means clustering performed very well, with 82.39% background noise removal.

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- The algorithm effectively differentiated ink strokes from handwritten signatures, removed seals, stamps, and printed text overlaps, producing cleaner and more consistent signature crops to match.

Fig. 7 demonstrates the output of signature cleaning model.

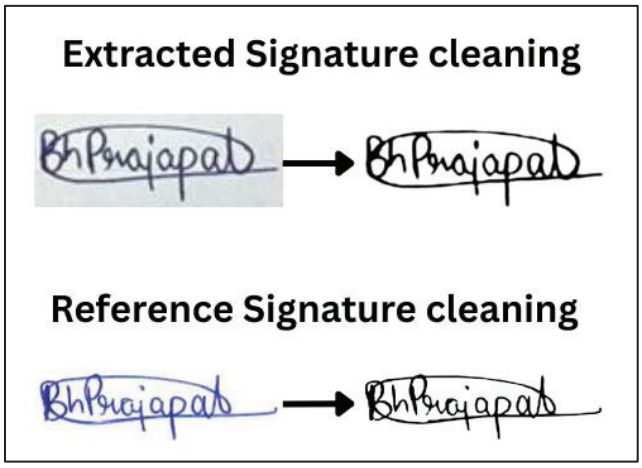


Fig. 7. Signature cleaning model's output

D. Signature Matching Performance:

- The Siamese network with VGG16 as the backbone performed better ROC AUC at 0.9843 (98.43%).

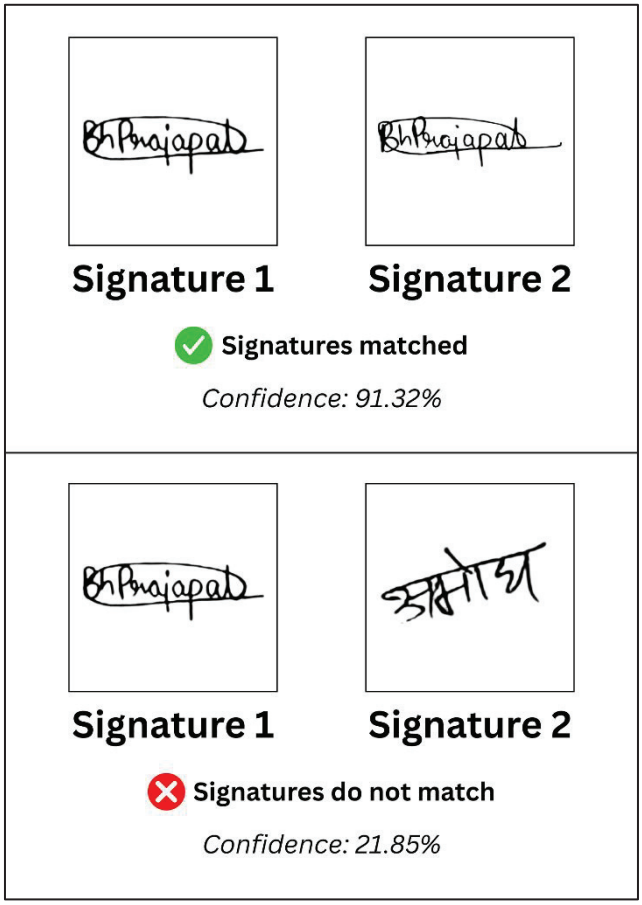


Fig. 8. Signature matching model's output

- This confirmed the system to be effective in discriminating real and forgery signatures and was stable within intra-writer variation and multilingual applicability.

Fig. 8 demonstrates the output of signature matching model.

E. Collective System Effectiveness:

- These cumulative outcomes recognize the system to be viable, stable, and deployable in actual multilingual document verification situations, particularly within banking, government, and legal contexts.

VII. RESULTS AND DISCUSSION

Systematic performance analysis was carried out to give a measure of each step's performance correlation. Quantitative findings of the most significant results of the modular system are presented in Table II.

TABLE II. PERFORMANCE METRICS OF THE THREE-PHASE SIGNATURE VERIFICATION SYSTEM

Phase	Metric	Achieved Value
Detection (YOLOv5)	mAP@0.5	92.71%
Cleaning (K-Means Clustering)	Noise Reduction	82.39%
Matching (Siamese VGG16)	ROC AUC	0.9843 (98.43%)

Cross-script testing validated consistent performance with Hindi, Marathi, and Urdu signatures, questioning the multilingual ability of the system. It was less robust with very cursive signature writing styles, with over-inking or poor scan quality, particularly for the case of the Urdu samples, where there was not enough ligature spacing. The samples are promising to further enhance with more advanced data augmentation and adaptive preprocessing.

Perceptual vision assisted YOLOv5 in detecting signature locations efficiently even remotely among dense document content. Perceptive vision was strongly augmented when background content was properly evaded during post-cleaning scanning when performed for the first time. Basic adoption by the K-Means algorithm was, however, in an indefensible position to denoise with or without tagged guidance or templating.

ROC curve analysis revealed ideal discrimination of actual and pseudo signature pairs. The almost ideal ROC AUC of 0.984 speaks volumes about the discriminative potential of the Siamese feature embeddings and efficiency of contrastive loss training. There were nearly no false positives and false negatives at test time, which signifies model consistency.

Besides, qualitative user experience on visual verification tasks indicated pre-cleaning signatures by the cleaning module as more human-readable to examiners, another evidence of greater value in real use. Overall, modularity not only yielded satisfactory quantitative results but also turned out to be usable in real use, a prerequisite for successful large-scale use in banking, jurisprudence, and administration.

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion:

- This paper presents a very efficient and computationally costly multilingual signature verification system with no real-world document noise and variability control. By breaking down the verification process into three independent and

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modular phases— detection, cleaning, and matching—the system is made more flexible, simpler to maintain, and more suitable for generalization to other scripts and documents.

- Detection module, in a hat tip to its self-adaptive YOLOv5 model, functioned magically well to detect Hindi, Marathi, Gujarati, and Urdu text signature pattern types at astounding levels of accuracy. To disinfect, unsupervised K-Means clustering was utilized to disinfect out handwritten ink streams from background noises effectively without human-annotated masks. Finally, the matching stage, built against a Siamese network built from an encoder VGG16 and trained on contrastive loss, worked exceedingly well to recognize real vs. forged signatures.
- Quantitative results validate the system performance, detection mAP@0.5 of 92.71%, background noise eradication at 82.39%, and signature match ROC AUC of 0.9843 (98.43%). Results validate the high real-world use of the system in banking, legal, and administration cases with security and trust being of major concern.

B. Future Work:

Promising as is the present system, there exist some areas to improve for future work:

- **Attention-Augmented Matching:** Adding attention blocks like the Convolutional Block Attention Module (CBAM) to the Siamese model enables the model to focus even more intensively on the most important parts of a signature, making even better matches.
- **Edge Optimization:** Optimization for future consumption to enable real-time on lower-end devices will be tackled by techniques like quantization, pruning, and low-resource-computation use like MobileNet or EfficientNet.
- **Extended Language Support:** Beyond existing support for the four languages, other languages like Bengali, Tamil, Telugu, and Kannada can also be supported through some more multilingual synthetic training data.
- **Hybrid Denoising Techniques:** Hybridizing the clustering algorithms like K-Means and denoising algorithms like U-Net based on deep learning can result in even cleaner signature extraction, particularly of bad degraded or noisy documents.
- **Interactive Dashboard Development:** There will be created an interactive web-based dashboard where there will be document upload functionality, visually mark signatures found, similarity scores view, and audit verification functionalities with explainable AI.
- **Explainable AI Integration:** Integration of explanation capability into the system will enable end-users to see why two signatures are similar or dissimilar, building user trust and regulation compliance in highly regulated sectors such as banking and government identification verification.

With all these extra improvements, the system is now set to become a stable enterprise-class solution capable of dealing with the increased demand for multi-lingual, secure, and scalable signature verification today.

ACKNOWLEDGEMENT

I express my sincere gratitude to Dr. Nilkamal More and Mr. Chirag Desai for the guidance and supervision they provided during the project. Their constant encouragement, technical advice, and support played an important role in determining the direction and conduct of this research. I further appreciate the facilities and academic environment offered by the Department of Information Technology (AI&DS) of K. J. Somaiya College of Engineering to complete this project successfully.

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